MIL-C-22912B(OS)
9 September 1970
Superseding
MIL-C-22912A(OS)
16 September 1966

MILITARY SPECIFICATION

CONTROLLER, STEERING

(For Torpedo Mark 37 Mod 2)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification establishes the requrements for the procurement of a steering controller, LD 539991. The steering controller is used to order directional changes in horizontal (azimuth) heading of a torpedo.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-S-20708/16A	Synchros, 60 and 400 Cycle; General Specification for
MIL-E-82590	Environmental Requirements and Tests (For Torpedo Mk 37 Components and Assemblies)

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

FSC 1355

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MIL-STD-147	Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base

DRAWING s

Naval Ordnance systems Command

LD 539991	Steering Controller Assembly
LD 620008	Shipping (Packing) Condition (for one Steering Controller)
DWG 2239924	Wiring Diagram

(Copies of specifications, standards, drawings, and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATIONS

Handbook H28	Screw-Thread Standards for the Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Description. The steering Controller, LD 539991, consits primarily of a step positioner mechanically coupled to a synchro control differential transmitter. Direct current pulses applied to the step positioner cause rotation of the step positioner and control transformer shafts in accurate increments of two degrees. The synchro control transformer accepts electrical signals from the synchro pick-off of a course gyro. The signal is modified by the angular mechanical position of the control transformer shaft so that the output of the control transformer may be used to order directional changes in horizontal (azimuth) heading.

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3.2 General requirements

3.2.1 Data required. No data is required by this specification (other than reports accompanying samples submitted for preproducticn and periodic production testing), or by applicable docments referenced in Section 2, unless specified in the contractor order (see 6.2).

3.2.2 Preproduction sample. Unless otherwise specified in the contract or order, preproduction samples of the steering controllers shall be manufactured using the methods and procedures proposed for the production lot. These units shall be tested as specified in Section 4 herein and are for the purpose of determining, prior to starting production, that the contractor's production methods are capable of producting steering controllers that comply with the technical requirements of the contract. See 4.3.1.

3.2.3 Materials. Unless otherwise specified in the contract, requisition or order, all materials and purchased parts entering into the final assembly of the steering controller shall confom strictly to the detailed requirements indicated on the drawings and specifications in LD 539991.

3.2.4 Finishes, coatings, construction, and dimensions. Finishes, coatings, constriction, and dimensions for the steering controller shall be strictly in accordance with the applicable drawings, specifications, and publications listed in LD 539991.

3.2.5 Moisture, fungus, and corrosion. The steering controller shall be rendered moisture, fungus, and corrosion resistant in accordance with detail requirements specified on the drawings and specifications In LD 539991.

3.2.6 Marking. Marking of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed in LD 539991.

3.2.7 Threads. Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook HZ8. The class or fit for threads shall be specified on the applicable drawings.

3.2.8 Conflicting requirements. Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation and clarification.

3.3 Performance and product characteristics. The steering controller shall meet the following performance and product characteristics:

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3.3.1 Insulation resistance. The insulation resistance of the steering controller shall be not less than ten megohms between any terminal and the chassis under any service or environmental condition.

3.3.2 Alignment. With the dial of the step positioner set on zero degrees, the control differential transmitter shall be at tis electrical zero.

3.3.3 Phase. With the control differential transmitter connected through a synchro control transmitter to a single phase, 400 Hz, 115 volt source of ac power, the electrical output from the control differential transmitter of the steering controller shall be out of phase with the power source when the dial of the steering controller is rotated In a clockwise direction (i.e., the dial reading decreases).

3.3.4 Operation. The steering controller shall step accurately in increments of two degrees in both clockwise and counterclockwise directions throughout complete rotation of the shafts when operated at 8 rate of six steps per second. There shall be no skipping of steps or binding of the shafts at any point of rotation.

3.3.5 Operating life. The steering controller shall meet the requirements of 3.3.1 through 3.3.4 upon completion of operation of 1.25,000 steps In each direction of rotation under the conditions described in 4.5.1.

3.4 Environmental requirements. The steering controller shall meet the other applicable requirements of this specification during or after the tests defined or specified in 4.4.

3.4.1 Vibration. The steering controller shall meet the following vibration requirements

3.4.1.1 Vibration, nonoperating. The steering controller shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.1.2.

3.4.1.2 Vibration, operating. The steering controller shall show no damage and shall meet all the performance requirements and product characteristics of this specification during and after the test-specified in 4.4.1.3.

3.4.2 Temperature and humidity

3.4.2.1 High temperature and humidity, nonoperating. The steering controller shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.2.1

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3.4.2.2 High temperature, operating. The steering controller shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.4.2.2.

3.4.2.3 Low temperature, nonoperating. The steering controller shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.2.3.

3.4.2.4 Low temperature operating. The steering controller shall be capable of meet ing all the performance requirements of this specification while undergoing the test specified in 4.4.2.4.

3.5 Cleanup. Prior to and after final assembly all steering controllers shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges as well as rosin flash shall be removed.

3.6 Workmanship. The steering controller, Including all parts and accessoriess, shall be constructed and finished in a manner to ensure that it complies with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, painting, riveting, machine screw assemblage, welding, brazing, and freedom of parts from burrs and sharp edges.

The standards of workmanship exhibited in any approved prepriduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specif ically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the applier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the steering controller shall be classified as follows:

(a) Preproduction inspection (paragraph 4.4)

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(b) Quality conformance inspection (paragraph 4.5)

(c) Periodic production inspection (paragraph 4.4)

4.3 Sampling

4.3.1 Reproduction. A preproduction sample of five steering controller, manufactured In accordance with 3.2.2, shall be subjected to the preproduction tests detailed in 4.4 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the steering controller by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Lot size. Unless otherwise specified in the contract or order, each shipment of units of product shall be a separate "inspection lot" and may differ from the quantity specified in the total order or contract. In any event, all units of a single "inspection lot" shall be made to the same design, with the same materials and manufactureing techniques, and shall be part of one continuous production run.

k. 3.3 Periodic production samples. From each succeeding lot or grand lot of 300 or less that have passed the quality conformance inspection of 4.5, a random sample of five steering controllers will be selected for submission to an activity designated by the procuring actitity to determine compliance of the samples with the requirements of the contract, specification, and drawings. The steering controllers shall be considered satisfactory if they meet all specification and pertinent drawings requirements and withstand the tests specified in 4.4 and 4.5. Failure of the periodic production sample to comply with these requirements will result in the rejection of the lot, as determined by the procuring actitity.

4.4 Reproduction and periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the quality conformance inspections detailed in 4.5, shall be subjected to the following tests and examinations. Certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the procuring actitity.

4.4.1 Vibration

4.4.1.1 Mounting. The steering controller, less pack or packaging, shall be mounted by means of its mounting holes and tested as specified in 4.4.1.2 and 4.4.1.3.

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4.4.1.2 Vibration, nonoperating. The steering controller shall be sujected to the "Vibration, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.4.1.1.

4.4.1.3 Vibration, operating. The steering controller shall be subjected to the "Vibration, operating" test of MIL-E-82590 and during and thereafter shall meet the requirements of 3.4.1.2.

4.4.2 Temperature and humidity

4.4.2.1 High temperature and humidity nonoperating. The steering controller shall be subjected to the "High temperatue and humidity, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.4.2.1.

4.4.2.2 High temperature, operating. The steering controller shall be sub jetted to the "High temperature, operating" test of MIL-E-82590 and, While at the test temperature, shall meet the requirements of 3.4.2 2₀

4.4.2.3 Low temperature, nonoperating. The steering controller shall be subjected to the "Low temperature, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3•4.2• 3.

4.4.2.4 Low temperature, operating. The steering controller shall be subjected to the "Low temperature, operating" test of MIL-E-82590 and, while at the test temperature, shall meet the requirements of 3.4.2. 4

4.4.3 Operating life. With the steering controller connected as In 4. 5. 4. 4, operate the unit by introducing command signals to the step positioner through the keying switch or by means of the interrupter or signal generator for not less than 125,000 steps in each direction of rotation, under the environmental conditions described in 4.5.1. At the c conclusion of the test, the steering controller shall meet the performan requirements and product characteristics of this specification.

4. 5 Quality conformance inspection. Each steering controller delivered under contract shall be subjected to the following tests and examinations. Failure to meet any requirement specified herein shall be considered cause for rejection.

4.5.1 Test conditions. Unless otherwise specified, the steering controller shall be subjected to quality conformance tests under the following conditions:

4.5.1.1 Temperature: Room ambient 65 degrees F to 95 degrees F (18 degrees Centigrade (C) to 35 degrees C).

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4.5.1.2 Altitude. Normal ground.

4.5.1,3 Vibration. None.

4.5.1.4 Humidity. Room ambient to 95 percent relative maximum.

4.5.2 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and shall provide personnel for performing all quality conformance tests. The test equipment shall be adequate in quantity and, unless definite requirements are specified, shall be of sufficient accuracy and quality to permit performance of the required quality conformance tests.

4.5.3 Test equipment. The following Items of test equipment are required to perform the quality conformance tests set forth in this specification.

4.5.3.1 A megohmmeter rated at 500 volts dc, with a midscale reading of not less than 10 megohms.

4.5.3.2 A 115 volt, 400 Hz, single-phase, ac power supply.

4.5.3.3 Synchro control transmitter, Type 15CXa.

4.5.3.4 Power supply, dc, minus 24 volt, 200 ma minimum.

4.5.3.5 An interrupter that will be off for l/12 of a second and on for 1/12 of a second. The combined transfer time and tolerance for each “on” period and each “off” period shall be no greater than four milliseconds.

4.5.3.6 A vacuum tube voltmeter capable of reading at least 100 volts on the highest scale, with the lowest scale capable of reading 3 millivolts maximum.

4.5.3.7 A phase meter capable of determining the electrical phase difference between two 400 Hz signals.

4.5.3.8 A single-pole, double-throw, center-off, toggle switch rated at not less than 3 amperes.

4.5.4 Test procedure

4.5.4.1 Insulation resistance. Using the meghmmeter, measure the insulation resistance between every terminal of P902, Drawing 2239924, and the mounting bracket or chassis. The reading shall conform to 3.3.1.

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4.5.4.2 Alignment. With the steering controller. completely assembled in Its mounting bracket, set the dial at zero degrees reading. with the vacuum tube voltmeter (millmeter), using the method specified in MIL-S-20708/16A for a synchro differential transmitter, determine that the differential transmitter is on electrical zero by rotating the dial one step in a clockwise direction and one step in a counterclockwise direction.

4.5.4.3 Phase. Connect terminals R1 and R2 of the synchro control transmitter to line 1 and line 2, respectively, of the 115 volt, 400 Hz ac power source. Connect terminals S1, S2, and S3 of the transmitter to S1, S2, and S3, respectively, of the synchro differential transmitter of the steering controller. Set the dial of the steering controller at zero degrees reading. Clamp the shaft of the transmitter rigidly to prevent rotation. Rotate the dial of the steering controller In a clockwise direction (i. e., decreasing dial reading). With the phase meter, determine that the voltage from terminal 4 to terminal 5 is out of phase with the voltage from line 1 to line 2.

4.5.4.4 Operation. Connect one terminal of the 24 volt dc power source to terminal 1 of the steering controller and connect the other terminal, through the interrupter, to terminals 2 and 3. Make contact between terminals 1 and 2 and observe that the step-positioner rotates in a clockwise direction (i. e., decreasing dial reading). Continue making and breaking contact for a total of 180 times and observe that the dial has returned to its initial starting position without skipping steps or without binding at any point of rotation. Repeat the foregoing, making contact between terminals 1 and 3, the resultant direction of rotation being counterclockwise.

4.5.4.5 Visual and dimensional examination. Each steering controller shall be examined to determine conformance with the requirements for workmanship and dimensions.

4.5.4.6 Packaging packing and marking. Packaging, packing, and marking shall comply with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The levels of preservation and packaging shall be as specified (see 6.2).

5.1.1 Level

5.1.1.1 Unit packaging. The steering controller shall be packaged in accordance with Method IID of MIL-P-116 and LD 620008.

5.1.2 Level B. Not applicable.

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5.1.3 Level C

5.1.3.1 Unit packaging. The steering controller shall be packaged in accordance with Method III of MIL-P-116.

5.2 Packing. Packing shall be Level A or Level C as specified (see 6.2).

5.2.1 Number of packages per pack. The number of unit packages per pack, for Level C, shall be as specified in the contract or order.

5.2.2 Level A

5.2.2.1 Unit pack. The steering controller shall be packed in accordance with LD 2000 and may be shipped individually in small lots.

5.2.2.2 Palletizing. When specified, Level A packs shall be palletized in accordance with MIL-STD-147, Load Type XIII (see 6.2).

5.2.3 Level B. Not applicable.

5.2.4 Level C. The steering controller shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving actitity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Marking

5.3.1 Normal markings Level A. Unit packs shall be marked in accordance with MIL-STD-129 and LD 620008.

5.3.2 Normal markings Level C. Unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

5.3.3 Serial numbers. Unit packages and shipping containers shall be marked with the serial number of the units contained within.

6. NOTES

6.1 Intended use. The steering controller is intended for use in Torpedo Mk 37 Mod 2.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Any exception to this specification.

(c) Required level of preservation and packaging (see 5.1).

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(d) Required level of packing, including palletizing (see 5.2).

(e) Assigned actitity far preproduction and periodic preduction inspections.

(f) Data required other than that specified in 3.2.1.

(g) That additional samples required because of failure of any unit(s) to pass the prescribed periodic production Inspections shall be privided and inspected at the expense of the supplier.

6.3 Criteria for use of proper level of preservation, packaging, and packing shall be as follows:

Level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under Indeterminate condi- t ions for redistribute ion anywhere.

Level C. This level shall be used only when It is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

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